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Altitudinal structuring of sand flies (Diptera: Psychodidae) in the High-Atlas mountains (Morocco) and its relation to the risk of leishmaniasis transmission

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Abstract

This paper presents the results of entomological surveys on phlebotomine sand flies (Diptera: Psychodidae) in the Haouz of Marrakech and High-Atlas mountains (Morocco). Sand flies were captured with sticky traps from 25 stations with altitudes ranging between 400 and 1400 m. A total of 2742 specimens belonging to nine phlebotomine species was collected, *Phlebotomus* (*Larroussius*) *perniciosus* Newstead being the predominant species. There was a remarkable difference in the diversity of the sand fly fauna among the altitudes. Two associations of sand fly faunas were determined, the first one in lower altitude and the second one in higher altitude. The significance of the predominant species at any altitude range was discussed in terms of the risk of transmission of leishmaniasis.

Keywords: Phlebotomine sand flies; Altitude; Diversity; Correlation; Leishmaniasis risk

1. Introduction

Phlebotomine sand flies (Diptera: Psychodidae) are the vectors for leishmaniasis as well as for arboviruses and bartonellosis. In Morocco, leishmaniasis are endemic and remain to be a great public health problem (Moroccan Ministry of Public Health (MPH), 2001). Visceral leishmaniasis (VL) is widespread in all the country and is more frequent in its northern part. This form of the disease is caused by *Leishmania* (*Leishmania*) *infantun* Nicolle and transmitted by species of the subgenus *Larroussius*. Dog is the common reservoir host. In pre-saharan area, *L. major* Yakimoff and Schokhor causes the zoonotic cutaneous leishmaniasis (ZCL) with *Meriones shawi grandis* as the main reservoir host and *Phlebotomus* (*Phlebotomus*) *papatasi* Scopoli as the vector (Rioux et al., 1982, 1986a,b). Cutaneous leishmaniasis caused by *Leishmania tropica* Wright, is wide spread in the semi-arid regions with *Phlebotomus* (*Paraphlebotomus*) *sergenti* Parrot as the vector (Guilvard et al., 1991; Pratlong et al., 1991). Domestic dogs were found infected with *L. tropica* in some rural areas (Dereure et al., 1991), but they are not regarded as an animal reservoir. The transmission cycle still considered to be anthroponotic (MPH, 2001).

Previous studies in Morocco (Rioux et al., 1984, 1997; Rioux, 2001; Rispail et al., 2002; Rioux and de La

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Rocque, 2003) showed that the distribution of sand flies was due, in great part, to the bioclimate. In the present work, we discuss the possible effect of the altitude on the diversity and distribution of sand flies. Indeed, the altitude is not an ecological factor by itself, but it can act on the sand flies distribution by the diversity of habitats, relief, and by the gradient on climate that it offers (gradient on temperature, pressure, precipitation). Some studies demonstrated the effect of the altitude, through the temperature, on the morphometric and/or the molecular characteristics of some sand fly species (Marcondes et al., 1999; Belen et al., 2004). Current findings in Morocco showed that the distribution of some sand fly species were different according to the altitude (Guernaoui et al., 2005a).

Furthermore, more systematic surveys are required considering that sand fly distributions could be affected by the climate change and by the variation of ecological conditions due to the changing land-use and settlements. In addition, earlier distribution records probably contain errors because of the misidentification between sympatric species of the subgenus *Larroussius*, especially *Phlebotomus perniciosus* Newstead and *Phlebotomus longicuspis* Nitzulescu (Pesson et al., 2004; Guernaoui et al., 2005a).

The current study reports the results of entomological surveys in the High-Atlas mountains, Morocco, with the aim of investigating and quantifying the possible effects of altitude on the sand fly populations in relation with the distribution of leishmaniasis. The possible effects of climate warming on the leishmaniasis risk in the area under study is discussed.

2. Materials and methods

2.1. Study area

Field studies were conducted, in July and September 2002, in the Haouz of Marrakech and in the High-Atlas mountains (Fig. 1), Morocco. We chose 25 stations with an altitude ranging between 400 and 1400 m above sea level. At least, four stations were prospected in each 200 m altitude range.

In these areas, the bioclimate is arid to semi-arid. The mean monthly temperature varies from 24 to 27 °C in July, and from 20 to 26 °C in September. The mean annual rainfall is low (less than 400 mm per year in the plain of Haouz and about 600 mm per year in the High-Atlas mountains). The vegetation is very degraded due to the human activities. In the plain of Haouz, it is composed mainly by *Zizyphus lotus* Lamarck, *Retama monosperma* Boiss, *Pistacia atlantica* Desf, *Acacia gummifera* Willd and the Chenopodiacae species (*Salicornia Arabica* L., *Atriplex* spp., *Haloxylon scoparium* Pomel). The slopes of the High-Atlas are wooded up to 1500 m by varied tree species (juniper tree, oak tree, thuja tree).

2.2. Sand flies sampling and monitoring

Sand flies were captured overnight with castor oil impregnated A4 papers (sticky traps). In each site, 30 traps were placed, on two consecutive nights. They were set in various biotopes; inside and around human dwelling and animal housing, close to the vegetation and crevices in walls.

Specimens were stored at 70°C ethanol. Before the dissection, they were cleared, dehydrated, and coloured with fuschin 3‰. They were mounted in Canada balsam.

Identification was made according to the shapes of the pharynges, cibarial teeth, female spermathecae and male genitalia. The differentiation between males of *P. perniciosus* and that of *P. longicuspis* was made by examining both the copulatory valves form and the number of coxite hairs (Pesson et al., 2004; Guernaoui et al., 2005a).

3. Results

In this entomological survey, a total of 2742 specimens belonging to nine phlebotomine species was collected. *P. perniciosus* (36.5%), *P. longicuspis* (15.1%), *P. papatasi* (13.3%), *P. sergenti* (12.5%), *Sergentomyia* (*Sergentomyia*) *fallax* Parrot (11.6%) and *S.* (*S.*) *minuta* Rondani (5.8%) were the most prevalent species. While *S.* (*Grassomyia*) *dreyfussi* Parrot (2.6%), *P.* (*L.*) *ariasi* Tonnoir (2.2%) and *P.* (*Paraphlebotomus*) *alexandri* Sinton (0.5%) were less frequent. The number and relative abundance of these species at different altitudinal ranges are given in Table 1.

There was a difference in the diversity of the sand fly fauna among the altitudes as indicated by the values of Shannon–Weiner index (*H*) (Table 2). From 800 m altitude, the diversity was more significant. The richness, diversity and evenness were maximal at 800–999 m altitude. All the nine phlebotomine species were present without the predominance of anyone. The relatively low evenness (0.65), even the high richness, at the altitude range of 1000–1199 m may be due to the quasi-dominance of *P. perniciosus* (40.9%) (Table 1).

The sand fly communities similarity result of Jacard's coefficient (I_{Jacard}) indicated that the sand fly communities similarity was highest between 800–999 and 1000–1199 m altitude ranges and lowest between 400–599 and 1200–1400 m altitude ranges. Between 800

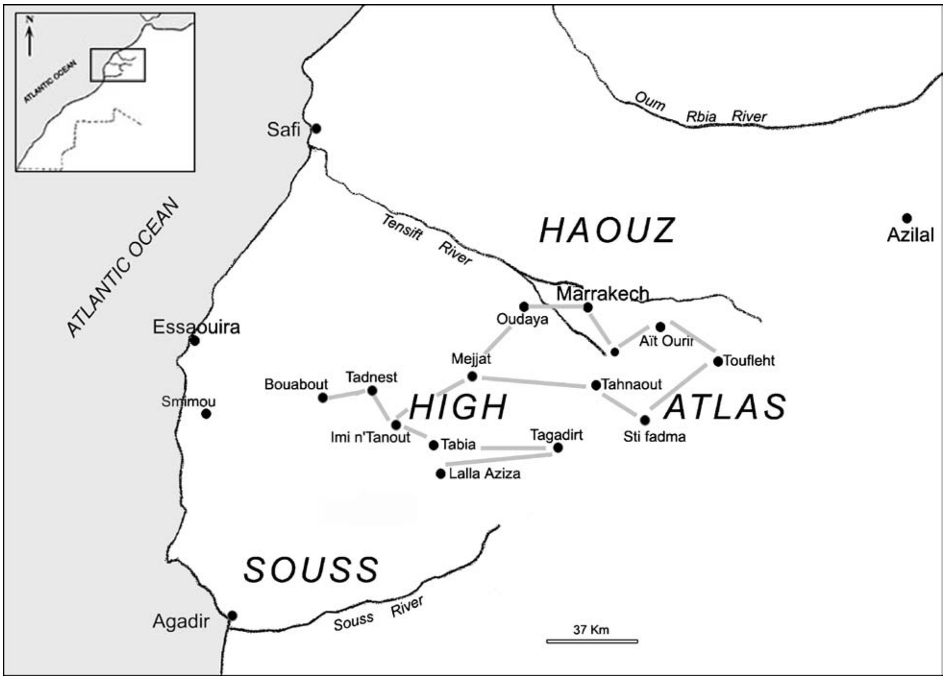

Fig. 1. Map of the study area showing the main sites sampled and the road taken (grey line).

and 1199 m altitude, 100% of sand fly communities were similar ($I_{\text{Jacard}} = 1$), while only 37.5% of sand fly communities at 400–599 and 1200–1400 m were the same ($I_{\text{Jacard}} = 0.375$). There seemed to be two altitude associated sand fly faunas, one in higher altitude and another one in the lower altitude. The transition between these two assemblage is about 800 m altitude, here, the species richness and diversity were the highest.

The altitudinal distributions of the most abundant *Phlebotomus* species are shown in Fig. 2. The species *P. sergenti* was collected at all altitudes and its distribution showed a low positive correlation with the altitude ($r = 0.16$). Its highest densities were registered at 800–999 m altitude. In contrast, *P. perniciosus* distribution was highly positively correlated with the altitude ($r = 0.42$). It was absent at 400–599 m and it reached a peak at 1000–1199 m altitude. On the other hand, the species *P. papatasi* showed a negative correlation with the altitude ($r = -0.82$). *P. papatasi* densities were very important in plain (400–599 m) and they were lower at

Table 1
Number (*N*) and relative abundance (%) of sand fly species collected at different altitude ranges of the High-Atlas mountains (Morocco)

Species	Altitudinal ranges (m)										Total (25 sites)	
	400–599 (5 sites)		600–799 (5 sites)		800–999 (6 sites)		1000–1199 (5 sites)		1200–1400 (4 sites)		*N*	%
	N	%	*N*	%	*N*	%	*N*	%	*N*	%		
P. papatasi	217	59.3	13	3.5	77	21	59	16.1	0	0	366	13.3
P. sergenti	3	0.8	14	4.1	160	46.8	154	45	11	3.2	342	12.5
P. alexandri	0	0	0	0	8	61.5	4	30.8	1	7.7	13	0.5
P. perniciosus	0	0	13	1.3	162	16.2	807	80.7	18	1.8	1000	36.5
P. longicuspis	2	0.5	273	66	47	11.3	86	20.8	6	1.5	414	15.1
P. ariasi	0	0	0	0	1	1.6	58	96.7	1	1.6	60	2.2
S. fallax	64	20.2	21	6.6	124	39.1	108	34.1	0	0	317	11.6
S. minuta	14	8.7	3	1.9	90	56.2	46	28.8	7	4.4	160	5.8
S. dreyfussi	0	0	5	7.2	22	31.4	43	61.4	0	0	70	2.6
Total	300	10.9	342	12.5	691	25.2	1365	49.8	44	1.6	2742	–

Table 2
The Shannon–Weiner diversity index (*H*), evenness (*E*) and richness (*S*) for the sand fly species at different altitude ranges of the High-Atlas mountains

Altitudes (m)	*H*	*E*	*S*
400–599	1.13	0.49	5
600–799	1.20	0.43	7
800–999	2.67	0.84	9
1000–1199	2.08	0.65	9
1200–1399	2.09	0.80	6

A higher value of *H* indicates greater diversity than a lower value. The evenness (*E*) is calculated by dividing *H* by the maximum value for a given population.

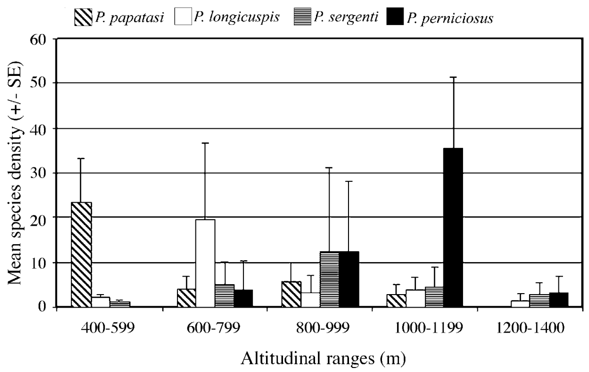

Fig. 2. Distribution of the most frequent *Phlebotomus* species at different altitudinal ranges (data are presented as the number of specimens per meter square of sticky trap per night ± standard error (S.E.)).

the other altitudes. It was absent from 1200 m altitude. *P. longicuspis* was collected at all altitudes with a peak at 600–799 m altitude.

4. Discussion

In the present study, six *Phlebotomus* and three *Sergentomyia* species were identified. All proven and suspected vectors of leishmaniasis in Morocco are present in the area under study. There was a remarkable difference in the diversity of the sand fly fauna among the altitudes. Two associations of sand fly faunas were determined, the first one in lower altitude and the second one in higher altitude. The species richness and diversity were the highest at about 800 m altitude which could correspond to the transition between the lowland and the mountain (ecotone).

P. papatasi, the proven vector of *L. major* in Morocco (Rioux et al., 1986a), was the dominant species in the arid plain of the Haouz of Marrakech. The high occurrence of *P. papatasi* in lowland and its low frequency in the mountain could be explained by its preference for the arid ans perarid areas. Moreover, its density increases with the aridity (Croset et al., 1974). Contrary to *P. papatasi*, *P. perniciosus* and *P. ariasi* – two proven vectors of *L. infantum* in the Mediterranean basin (Rioux et al., 1986c) – prefer sub-humid and humid bioclimates (Rioux et al., 1984). They were collected in the mountain and they were scare or absent in the lowland which could be related with the heightening of the aridity in the area under study. In fact, geographic repartition of vectors could change due to the climate warming (moving toward the northern regions or higher altitudes) (Rioux et al., 1997; Rhodain, 2000; Rioux and de La Rocque, 2003). The stations where we collected *P. ariasi* have an altitude ranging between 1000 and 1400 m. Furthermore, Dereure et al. (1986) had noted the absence of *P. ariasi* at altitudes lower than 1000 m in the northern slope of the High-Atlas. They cited *P. pernicious* as a scare species which could be caused by the confusion between *P. longicuspis* and the atypical form of *P. perniciosus*. Moreover, in this region, the male of *P. perniciosus* does not display the bifid copulatory valves characteristic of the species, but rather, copulatory valves curved at their apex with 8–19 coxite hairs (Guernaoui et al., 2005a).

P. longicuspis is a suspected vector of *L. infantum* (Killick Kendrick, 1990). It was collected at all altitudes but it was more abundant between 600 and 799 m altitude. In Spain, only *P. ariasi* was collected at an altitude >750 m a.s.l. and the distribution of canine leishmaniasis coincided with the regions where *P. ariasi* and/or *P. perniciosus* were present (Aransay et al., 2004). In the High and Anti-Atlas mountains (Morocco), Dereure et al. (1986) demonstrated the presence of an autochthonous canine visceral leishmaniasis cycle. Three cases were identified in three villages at an altitude higher than 1100 m. Infantile VL occurs sporadically (MPH, 2001). In these areas, the possible role of both *P. perniciosus* and *P. ariasi* in the *L. infantum* transmission cycle is to be considered. *P. sergenti*, the proven vector of *L. tropica* in Morocco (Pratlong et al., 1991), was recently suspected to be the vector in Chichaoua focus of CL (Guernaoui et al., 2005b). In this study, *P. sergenti* was widespread. It was more abundant between 800 and 999 m altitude but no significant correlation was found between its density and the altitude. Furthermore, even if this vector preferred the semi-arid bioclimate, it was collected in all bioclimatic belts and was largely widespread in the whole of Morocco (Bailly-Choumara et al., 1971; Rioux et al., 1984).

The temperature is one of the main factors preventing the spread of both visceral and cutaneous leishmaniasis (Kuhn, 1999) to Northern Europe. This ecological factor vary with the altitude according to the thermal altitudinal gradient (−0.6 °C per 100 m). The possible relationship between the leishmaniasis transmission and altitude may be closely related to many factors as the temperature suitable for the evolution of *Leishmania* in the sand flies (Rioux et al., 1985). This study showed that altitude has an influence upon the spatial distribution and density of the sand fly vectors. Cross and Hyams (1996) predicted that raising the ambient temperature by 1–5 °C due to global warming would result in the expansion of the geographical distribution of *P. papatasi* in Southwest Asia and increase the transmission season of *L. major*. In this study, due to the aridity, *P. papatasi* densities were very important in lowland. In addition, it was active throughout the year in the urban area of Marrakech (Boussaa et al., 2005). This suggests the necessity to prevent the risk of extension of *L. major* foci from southern Morocco as it was predicted by Rioux et al. (1997) and Rioux and de La Rocque (2003). The wide distribution of *P. sergenti* corroborates with that of ACL and could increase the risk to spread *L. tropica* from Chichaoua focus to Marrakech city and other nonendemic sites in the High-Atlas mountains. *P. perniciosus* was largely widespread while *P. ariasi* was less frequent. Further investigations are needed to clarify the *L. infantum* transmission risk in this area by isolating and typing *Leishmania* strains from vectors.

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